TeenAstro user manual

Other languages: français · Deutsch

A field manual for the mount user. It describes the hand controller, the main unit, tracking, goto, alignment, and the mechanical settings as they exist in the current firmware.

It follows the path of the Astro-Electronic FS2 manual (connections, menus, changing a value, rotation direction, speeds, goto, German mount, encoders, visitor mode), and it is based on the hand-controller menus, the main unit, and the TeenAstro group wiki.

You do not need to read all of it before the first night. Chapters 3, 5, 7, and 8 are enough to start. A night on a mount that is already set up is chapter 26. The rest covers mechanical setup, a full alignment, and troubleshooting.

Menu names in bold are the labels of the English hand-controller firmware.

1. What the system is

TeenAstro is a controller for equatorial and alt-azimuth mounts. It was built by an FS2 user who wanted an open successor: the same night-time actions (hand controller, tracking, goto, sync), with stepper motors, a two-star alignment model, and a link to a computer.

Part	Role
Main unit	Computes the sky, drives the motors, stores the mount, the time, and the site
Hand controller (SHC)	Display, seven buttons, catalogs, settings
Wi-Fi interface	Bridge between the main unit and a phone or a computer
Focuser	Focus motor, optional
TeenAstro app	Dashboard, planetarium, goto
ASCOM driver	Stellarium, Cartes du Ciel, and other ASCOM software

The main unit is the only place that holds the mount position. The hand controller, Wi-Fi, and the computer send it commands and display what it answers.

Two mount parameter sets can coexist (mount 0 and mount 1). You switch sets in the Mount menu. A reboot is then required.

2. Precautions

- Do not connect or disconnect motor cables while the system is powered. Disconnecting a stepper under power can damage the driver.
- Before you change the mount type, the gear ratio, the step count, or the rotation direction, write down the values shown by **Show Settings**, or save them with TeenAstroConfig. A wrong parameter set turns a working mount into one that no longer points.
- The **Home** position at power-on is the mechanical reference. If the tube is not there, the displayed coordinates are wrong until a sync or a new alignment.
- **Visitor** mode hides the settings. Switch back to **Administrator** before a configuration session.
- A factory reset erases time, site, mechanics, and alignment. It asks for confirmation.

Supply voltage, connectors, and the schematic depend on the board (classic unit, Mini, Tiny). Wiring for each board is described in the group wiki, not in this manual.

3. The hand controller

Seven buttons: **Shift**, **North**, **South**, **East**, **West**, **F**, and **f**.

Outside a menu, North, South, East, and West move the tube. The current speed is shown by an icon at the top of the display.

Gesture	Effect
Short press on Shift	Next display page
Shift held + East	**Telescope Action**
Shift held + North	**Set Speed**
Shift held + South	**Display** (turn off, contrast)
Shift held + West	**Telescope Settings**
Shift held + F	**Focuser Settings**
Shift held + f	**Focuser Action**

Inside a menu:

Button	Effect
North / South	Move up or down the list
West	Enter the line, or increase a value
East	Go back, or decrease a value
F	Confirm
f or Shift	Leave the menu

Ergonomics turns the hand controller over for a left-hander: the display rotates 180° and the North/South and East/West buttons are swapped. The change takes effect after a reboot.

Button Speed (Slow, Medium, Fast) sets how fast a held key repeats. It does not set the mount speed.

The display dims, then sleeps, after a delay. **SHC Settings → Display** sets sleep, deep sleep, and the OLED sub-model. A press on Shift wakes the display. **Shift + South** can also turn it off at once or change the contrast (Min, Low, High, Max).

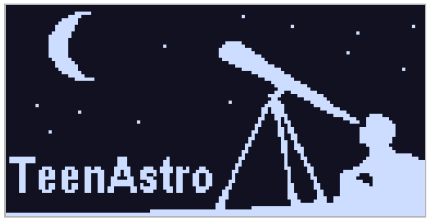
---

4. Reading the display

The pictures were taken from the hand-controller emulator, French firmware 1.6.7. The real display is 128×64 pixels; here it is scaled four times. An English or German build has the same layout. Only the words change, so the French text on these pictures is the French firmware, not a different screen.

At power on

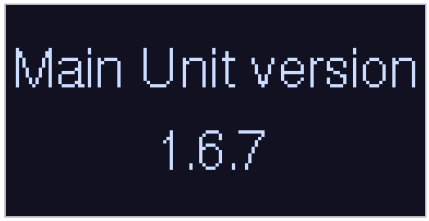
The hand controller shows the logo, then the versions, the board and the drivers, the focuser if it answers, and finally the time if no GNSS is connected.



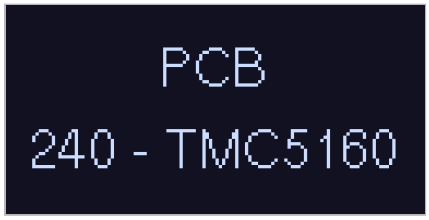
TeenAstro logo



Hand-controller version



Main-unit version



Board and drivers



Focuser version



Time and date at startup

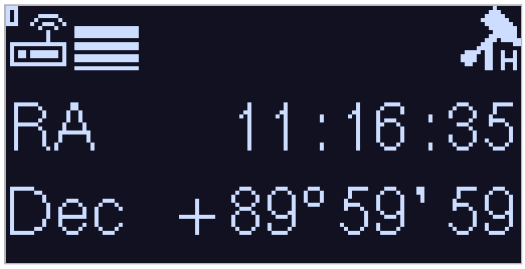
If the hand controller shows **ERROR** and **version**, its firmware and the main-unit firmware do not match. Reflash both from the same release.

If it shows **Not Connected**, the serial link to the main unit is lost. The hand controller reboots. Check the hand-controller cable before looking further.

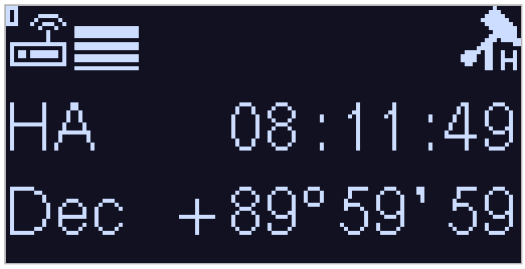
Pages

A short press on Shift goes to the next page.
`libraries/TeenAstoCustomizations/TeenAstoCustomizations.h` decides which pages exist. As shipped, four are on: right ascension and declination, azimuth and altitude, time, and the focuser. The others appear when the matching line is uncommented (`HA_PAGE`, `PUSH_PAGE`, `AXIS_STEP_PAGE`, `AXIS_DEG_PAGE`). During an alignment, the alignment screen replaces the current page.

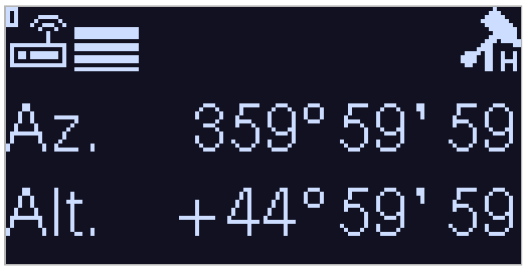
At home on an equatorial mount the declination is at the pole and the home icon is on.



Right ascension and declination
Default page. **RA** is right ascension, **Dec** is declination. Top left: Wi-Fi not linked, then the manual speed (here two bars, slow). Top right: home.



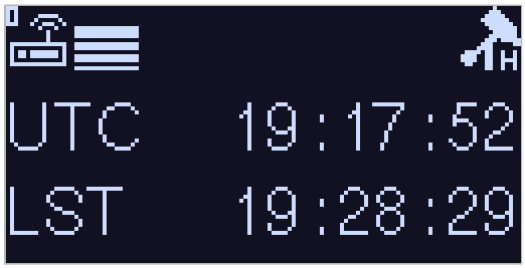
Hour angle
Optional page. **HA** is the hour angle, **Dec** the declination. It shows which side of the meridian the object is on.



Azimuth and altitude
Default page. **Az.** is azimuth, **Alt.** is altitude. At an equatorial home the tube points at the pole, so the altitude depends on the site latitude.



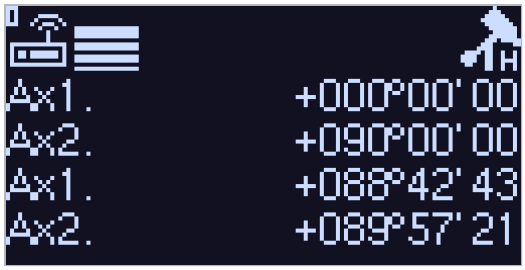
Push-to page
Optional page. Without encoders it stays empty: only the status row is drawn. With encoders enabled it shows the distance left on each axis to reach the target.



Time
Default page. **UTC** is universal time, **LST** is local sidereal time.



Axes in steps
Optional page. The two numbers are the step counters of axis 1, then axis 2. They show that a motor is turning. They are not sky coordinates.



Axes in degrees
Optional page. The first pair is the motor angle of axes 1 and 2. The second pair is the same angles after correction (backlash, model). Without encoders all four lines are labelled Ax1 and Ax2.



Focuser
Default page when a focuser answers. **F Position** is the focus position, **F Speed** its speed. If it does not answer, the page shows **Focuser** and **Not Connected**.

Menus

A menu opens by holding Shift, then pressing a direction. Shift or **f** leaves. East goes back. West or **F** enters the highlighted line. North and South move the highlight.

The words below are the French firmware. In English they are **Telescope Action**, **Set Speed**, **Display**, **Telescope Settings**, and **Hand Controller**.



Telescope action, Shift and East
Goto, **Sync**, and **Align** are the first lines. The list continues with the gear check, tracking, the side of pier, saving the coordinates, the lock, and the spiral. A parked mount offers only unpark.



Set speed, Shift and North
The highlighted line is the speed in use. The five steps are Guiding, Slow, Medium, Fast, and Max. The bars at the left of the main screen follow that choice: one bar for guiding, then two, three, four, and five for Max.



Display, Shift and South
Turn off blanks the display at once. **Contrast** offers Min, Low, High, and Max. The OLED font does not have every accent, so a French title can lose its accent.



Telescope settings, Shift and West
In administrator mode: **Hand Controller**, **Time & Site**, **Park & Home**, then Mount, the main-unit information, and Wi-Fi. In visitor mode this menu contains only **Rights**.



SHC Settings
The title of this screen is **SHC Settings**. **Rights** chooses administrator or visitor. **Display** sets the dim delay, the deep sleep, and the OLED model. **Button Speed** sets how fast a held key repeats (slow, medium, fast). It does not set the mount speed. Ergonomics and the hand-controller reset follow.

Icons

The status row is a line of 16×16 symbols. On the left: Wi-Fi, then the speed if the motors are powered, then the Shift arrow while that key is held. On the right: the mount state. Tracking, slewing, home, and park share one slot: the highest priority wins. Errors are added beside it.

Wi-Fi. A filled case means linked. An outline means the interface is on but has no link. The four drawings are the three station profiles (0, 1, 2) and the access point.

Linked	Not linked	Role
		Station, profile 0
		Station, profile 1
		Station, profile 2
		Access point

Manual speed. The number of bars is the step chosen in Set Speed. The icon is drawn only while the motors are powered.

	Step
	Guiding
	Slow
	Medium
	Fast
	Max, the goto speed

Where the tube is.

	Meaning
	At the home position
	Parked
	Parking in progress
	Parking failed
	Tracking stopped (two vertical bars)
	Tracking on, rate not shown (triangle)
	Sidereal tracking
	Solar tracking
	Lunar tracking

	Meaning
	Tracking at the object's rate (comet, drift)
	Tracking with correction on right ascension only
	Tracking with correction on both axes
	Equatorial goto
	Alt-azimuth goto
	Meridian flip
	A slew is in progress, type not shown
	Pier side east
	Pier side west

The tracking triangle is overlaid with the star, the sun, the moon, or the target, then with a 1 or a 2 when a tracking correction is active. During a goto that triangle is replaced by the slew icon.

Alignment, guiding, spiral.

	Meaning
	Alignment, star 1
	Alignment, star 2
	Alignment, star 3 or later
	An alignment model is stored
	Spiral search running
	Frame of a pulse-guide move
	Pulse toward north
	Pulse toward south
	Pulse toward east
	Pulse toward west
	Recentering (bullseye)
	Guiding at the track rate (crosshair)

The guide arrows are drawn on top of the frame, the bullseye, or the crosshair, depending on whether the mount is receiving an ST-4 pulse, a recenter, or a guide at the track rate.

The letters **eF** appear in the status row when an eFinder is connected.

GNSS, lock, Shift. The GNSS icon appears only at home or at park.

	Meaning
	Time and location synchronized
	Time only
	Location only
	Shift is held
	Locked, without saying which
	Telescope locked
	Focuser locked
	Telescope and focuser locked

Errors. They replace the normal state on the right, or they are added next to it. The short text is what the icon itself draws.

	Meaning
	Axis 1 limit
	Axis 2 limit
	Below the horizon
	Meridian limit
	Under the pole
	Pier side this mount type does not use (axis 2 past 90°). Gotos, slews and tracking stop; move back by hand or with a goto
	Motor fault

5. How to change a value

The rule is the same everywhere, as on the FS2.

1. Open the menu (usually Shift + West for settings, Shift + East for night-time actions).
2. Move down with South to the line, then enter it with West or F.
3. For a list (mount type, direction, microsteps), North and South move the highlight, F stores it.
4. For a number (gear, backlash, latitude), West increases, East decreases, F stores it. The step size is fixed for that screen.
5. **Set** confirms the write. **ERROR**, or a failed command, means the value was not stored.
6. East or f goes up one level without changing the value being edited.

Some changes require a reboot: mount type, choice of mount 0 or 1, enabling motors or encoders, ergonomics, Wi-Fi. The display then shows **Press key** / **to reboot**. Press a key and let the system restart. Do not cut power in the middle of that message.

6. First power-on

At the first power-on, the main unit assumes it is at the home position. That is not yet a mount aligned on the sky. It is only the mechanical origin.

What to set before you point

1. **Mount type**: German Equatorial, Equatorial Fork, Altazimutal, or Altazimutal Fork.
2. The **gear**, the **steps per rotation**, and the **microsteps** of each motor (chapter 9).
3. The **rotation direction** (chapter 8).
4. The **site** (latitude, longitude, elevation) and the **time** (chapter 10).

Without site and time, sidereal tracking can still run, but equatorial coordinates and the catalogs do not match the sky.

Default home position

- Equatorial mount: tube toward the celestial pole. On a German mount, the counterweight bar is down.
- Alt-azimuth mount: altitude 0° (tube horizontal) and azimuth 180° (tube toward the south). The base must be level.

Place the tube in that position **before** you switch on, or switch on, drive it home with the buttons, and store the position (chapter 11).

You can skip home for a first try: move the tube by hand onto an obvious object (Moon, planet, bright star), do a **Sync**, then use goto. A two-star alignment, however, starts from the home position, or from a star whose pier side you set.

Short check suggested by the group

1. Goto **Home**, then, with the clutches loose if needed, point the tube at the pole.
2. Goto a bright star, center it, do a **Sync**.

That is enough for a visual night if the mechanical polar alignment is already good. For a model that absorbs a polar-alignment error, use the alignment in chapter 15.

---

7. Mount type, refraction, pole

Telescope Settings → Mount

Line	Content
Mount	Selects parameter set 0 or 1. Reboot
Mount type	German Equatorial, Equatorial Fork, Altazimutal, Altazimutal Fork. Reboot
Motors	Mechanics, speeds, tracking
Encoders	Position encoders
Limits	Horizon, zenith, axes, meridian
Mount error	CH and NP held in a two-star alignment
Refraction	See below
Reticule	Polar-finder brightness, if that output is wired

Mount error: **Off** or **On**. When **On**, enter **CH**, then **NP**. Each value is in degrees, from −5 to +5, with three decimals, in Wallace's sign.

Leave it **Off** for the usual two-star alignment. Turn it **On** when CH and NP are already known, from a previous **4 Stars** or **3+3 Stars** alignment, or from a measurement of the mechanics. The signs are Wallace's TPOINT signs: CH is collimation and NP is non-perpendicularity. A two-star alignment then holds both numbers and still measures the pole. The start screen says **Measures pole**. If both numbers are zero, nothing is held and the two-star alignment stays the usual one. **4 Stars**, **3+3 Stars**, and **2 Stars Mech.** ignore this setting.

The same item is the last line of **Telescope Action → Align**.

On the hand controller web page the card is labelled **Mount error** and sits above refraction. **Off** or **On** comes first; the text after the choice is “Hold CH and NP”. CH and NP are each their own row: the value, **Upload**, then the name and the unit, in degrees, ±5, in Wallace's sign. The web page is in English on every firmware.

Refraction → Goto: ON or OFF. When on, atmospheric refraction is included in the slew (Saemundsson on the way out, Bennett on the way back).

Refraction → Polar Alignment (equatorial mounts only):

- **Apparent Pole** if you use a polar finder, which shows the pole where refraction places it.
- **True Pole** if the polar alignment aims at the true polar axis, already corrected for refraction.

On an alt-azimuth mount, only goto refraction is offered: there is no mechanical pole to choose.

---

8. Rotation direction

Test at slow or medium speed, tube roughly at home, looking through the finder or an eyepiece.

German equatorial

- South must send the tube west. If it goes east, reverse axis 1, then check again.
- East must send the tube east. If not, reverse the axis that is wrong and check again.

Equatorial fork

- South must send the tube south. If not, reverse axis 2.
- East must send the tube east. If not, reverse axis 1.

Alt-azimuth: East and West command azimuth (axis 1), North and South command altitude (axis 2). The tube must move in the direction named by the button.

The setting is **Mount** → **Motors** → **Motor 1** or **Motor 2** → **Rotation**: **Direct** or **Reverse**.

Axis 1 is right ascension or azimuth. Axis 2 is declination or altitude.

---

9. Motors, gear, current, backlash

Mount → **Motors**, with motors enabled:

Show Settings, Motor 1, Motor 2, Acceleration, Speed, Tracking, Settle Time, Disable.

Show Settings recalls, for each axis, the direction, the gear ratio, the steps per rotation, the microsteps, the backlash in arcseconds, and the low and high currents. Walk through those screens and write the values down before you change anything.

Gear

This is the number of motor revolutions for one revolution of the axis, teeth included.

Example: a 360-tooth wheel and a worm, with no other reduction: gear 360. A 2:1 pulley ahead of the worm makes the gear 720. A 5:1 planetary reducer on the same axis makes it 1800.

The accepted value runs from 1 to 60000, with three decimal places. Enter the total ratio, not only the wheel tooth count, if another stage exists.

Gear Check (chapter 19) measures this ratio on the sky and offers a corrected value.

Steps per rotation and microsteps

Steps per Rot. is the number of full steps of the motor, usually 200 or 400. The entry runs from 20 to 400.

MicroStep: 2, 4, 8, 16 (~256), 32, 64, 128, 256. The group most often keeps **16 (~256)** on TMC drivers: 16 real microsteps, with internal interpolation to 256. Going past 16 without a reason cuts the maximum slew speed more than it improves tracking.

Current

Low Curr. is for slow speeds. **High Curr.** is for fast slews. Entry is in steps of 100 mA peak, from 200 mA to 2800 mA.

Raise the current if the motor stalls on a goto or growls under load. Lower it if the motor heats with no useful load. High current must stay within the motor and driver ratings.

Backlash

Backlash is the compensation applied when the direction reverses, in arcseconds, from 0 to 999.

Backlash Speed sets how fast that backlash is taken up, from 16 to 64.

Set backlash only after the gear ratio and the direction are correct. Too much backlash makes the star jump at every reversal. Too little leaves a visible offset when you come back with the buttons.

Silent mode

Silent ON uses the driver's quiet mode (StealthChop on TMC). OFF favors torque at high speed. If a goto stalls only in Silent, switch it OFF for the tests, raise the high current, then try Silent again.

Acceleration

Acceleration is the distance, in degrees, over which the mount reaches maximum slew speed. From 0.1° to 25°. Too small a value shakes the mount. Too large a value makes every goto longer.

Settle time

Settle Time is the pause at the end of a slew, before tracking resumes. It lets vibrations die down.

Disabling the motors

Disable turns motor drive off and asks for a reboot. Useful when you want to move the tube by hand without fighting the holding torque. **Enable** turns them back on. While the motors are off, the menu is reduced to that single line.

---

10. Time, site, GNSS

Telescope Settings → Time & Site

Time

- **Clock**: civil time on the display.
- **Time Zone**: offset from UTC, daylight saving included. In Central Europe: +1 h in winter, +2 h in summer.
- **Date**
- **GNSS Time**: time read from the satellite receiver, if one is fitted.

The main unit derives local sidereal time from this. A clock wrong by one minute shifts pointing by about 15' in right ascension.

Site

Three sites can be stored.

- **Latitude**: positive in the north, negative in the south.
- **Longitude**: the sign convention is the one on the entry screen. Check the sign on a known object after the first sync. A reversed longitude is obvious: every goto lands on the opposite side in azimuth.
- **Site Elevation**: site elevation in meters. It enters the refraction calculation.
- **Select Site**: the active site.

GNSS sync

Sync to GNSS copies the receiver's time and location into the main unit. With no antenna, or indoors, the display shows **NO GNSS**. An automatic sync is offered when the mount is at home or parked and the receiver has a fix.

---

11. Home and park

The **home position** is the origin of the axes. The **park position** is where you store the tube at the end of the night, with the motors still aware of their angle, so the sky is found again at the next power-on without a new polar alignment.

	Home	Park
Role	Mechanical origin, start of alignment	Storage
Equatorial	Toward the pole, counterweight down on a German mount	Wherever the tube is safe
Default alt-azimuth	Altitude 0°, azimuth 180° (south)	Your choice

Telescope Settings → Park & Home

- **Set Park**: the current position becomes park.
- **Set Home**: the current position becomes home. Use this only when the tube really is in the pose you want as the origin.
- **Reset Home**: returns to the factory home of the mount type.

Telescope Action → Goto → Home or **Park** sends the tube there. **Sync → Home** or **Park** declares that the tube is already there, without moving it. Reserve that for the case where you placed it by hand.

When the mount is parked, the action menu offers only **Unpark**. Unpark before any move.

The home or park icon on the right of the display confirms the state. If it is missing while the tube looks like it is at home, the stored position and the real position have diverged: either bring the tube back, or define home again.

12. Speeds

Two different places:

- **Shift + North → Set Speed** chooses the button speed for the session: Guiding, Slow, Medium, Fast, Max. The highlighted line is the active step. The bars are explained in chapter 4. The picture shows the French firmware (**Réglage Vitesse**).
- **Motors → Speed** defines what those five steps mean, plus the speed selected at power-on.



Set Speed

Step	Setting	Range
Guiding	fraction of sidereal rate	0.10× to 1.00×
Slow, Medium, Fast	multiple of sidereal rate	1× to 255×
Max	goto speed	60× to 3600×, in steps of 60
Default Speed	step active at power-on	one of the five

Guiding speed is also the speed of the ST-4 autoguider port.

To center a star: Fast or Medium to bring it into the finder, Slow in the eyepiece, Guiding to place it in the center without overshooting.

13. Tracking

Telescope Action → Tracking

If tracking is stopped, the only line is **Start Tracking**.

If it is running:

- **Stop Tracking**
- **Sidereal**: stars
- **Lunar**: the Moon
- **Solar**: the Sun
- **Target**: the stored rate for a comet or a drifting object

The rate cannot be changed during a slew. The display reports **Currently Tracking** / **cannot be changed**.

Drift (comets, slow objects)

Motors → Tracking → Drift Speed

- **Right Asc.**: in seconds of time per sidereal interval (displayed unit `s/SI`), from −2 to +2.
- **Declination**: in arcseconds per sidereal interval (`"/SI`), from −2 to +2.

Set both from the ephemeris, start tracking, then choose **Target**. Switch back to Sidereal for the stars. Zero drift on both axes together with the Target rate returns to sidereal tracking.

Refraction on tracking

Motors → Tracking → Refraction: ON or OFF. Separate from goto refraction. It corrects the tracking rate for the object's altitude.

Tracking correction

Motors → Tracking → Tracking Corr., on equatorial mounts:

- **Right Asc.**: alignment and refraction correction act only on the hour-angle axis.
- **Both**: both axes are corrected.

On an alt-azimuth mount the correction is always on both axes, and this line does not appear. The tracking icon changes when correction is armed (one axis, or both).

The group recommends correction on both axes once a two-star alignment has been stored, and on right ascension alone if the polar alignment is already good and you do not want declination drift introduced by the model.

---

14. Goto, sync, catalogs

Shift + East → Telescope Action, mount unparked.

Line	Role
Goto	Slews the tube to the target
Pushto	Shows the direction, without motors, if encoders are enabled
Sync	Declares that the chosen object is in the center of the eyepiece
Align	Two-star model
Gear Check	Measures the gear ratio
Tracking	Starts, stops, selects the rate
Side of Pier	Sets the side, German mount
Save RADEC	Stores the current position as the user target
Lock	Blocks actions until Unlock
Spiral	Spiral search

The **Goto** menu and the **Sync** menu offer:

Catalogs, Solar System, Coordinates, User Defined, Home, Park.

Goto adds **Flip**.

Push-to offers Catalogs, Solar System, Coordinates, and User Defined, and only if encoders are enabled. Otherwise the display shows **Encoders** / **Not Connected**.

Solar system

Sun, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune, Moon. Positions are apparent, for the current site and time.

Point at the Sun only with a suitable full-aperture filter. The controller does not know whether a filter is in place.

Coordinates

- **J2000**: right ascension and declination of equinox 2000.0, as in most printed catalogs.
- **JNow**: apparent coordinates of the date.
- **Alt Az**: azimuth and altitude.
- **N, S, E, W**: the four cardinal points on the horizon (altitude 0°). Useful to check direction and level on an alt-azimuth mount.

The main unit applies precession, nutation, and aberration between J2000 and the date, then the equatorial-to-horizontal transform from latitude and sidereal time.

Catalogs

The list depends on the firmware build loaded in the hand controller. In practice it includes the bright stars, Messier, and, depending on the build, NGC, IC, Caldwell, Herschel, Collinder, doubles (STF, STT), and variables (GCVS). The English firmware names the bright-star catalog "Bright Stars".

Filters

Goto → Catalogs → Filters. The screen title is **FiltersAllow**. A **+** on both sides of a line means that filter is selected. **Reset Filters** clears them all.

Line	What it keeps
Above Horizon	The minimum altitude. **Filter Horizon** offers **Above Horizon**, then **Above 10 deg.**, **Above 20 deg.**, and so on up to **Above 70 deg.**
Constellation	One constellation. **Filter by Con** starts with **All**, then the abbreviations: Her for Hercules, Lyr for Lyra, And for Andromeda, Ori for Orion
Type	One deep-sky type. **Filter by Type** starts with **All**. The names are those of the database: Galaxy, Globular Clstr, Planetary Nebula, Open Cluster, Nebula
Magnitude	The faintest magnitude still shown. **Filter Magnitude** offers **All**, then 10, 11, 12, 13, 14, 15, and 16

Objects already below the horizon are omitted. **Above Horizon** therefore changes little. **Above 30 deg.** keeps only targets high enough for a comfortable eyepiece, clear of trees and poor seeing.

Magnitude runs opposite to intuition: a small number is a bright object. **10** drops everything of magnitude 10 and fainter. **All** cuts nothing. Under a bright sky with a small telescope, 10 or 11 is enough. 16 is only useful if the catalog and the sky support it.

Type changes the list only in a deep-sky catalog (Messier, NGC, and the others). On Bright Stars it removes nothing.

If the hand controller was built with the double-star or variable catalogs, two further lines appear:

- **Dbl* Min Sep.** and **Dbl* Max Sep.**: angular separation, from 0.2" to 100". The minimum must stay below the maximum, or the screen shows **Min Sep must / be < Max Sep.**
- **Var* Max Per.**: maximum period, from 0.5 day to 100 days. A variable with a longer period is hidden. **Off** removes that filter.

When a filter actually removes objects, the catalog title is wrapped in exclamation marks, for example **!Goto Messier!**. If nothing remains, the screen says **No Object**: loosen one step, or **Reset Filters**.

North and South scroll the objects that remain. F starts the goto or the sync, depending on the menu you came from. A full example, M13 then M31, is in chapter 26.

Center, then sync

1. Goto the object.
2. At the eyepiece, bring the object to the center with the buttons, slow speed then guiding.
3. **Sync** on the same object.

Sync recenters the position. It does not build the alignment model: a single object leaves the error in place as soon as you move away. For a model, use **Align**.

Save RADEC keeps the current coordinates. **Goto → User Defined** returns there.

Spiral

Spiral asks for a field of view, from 1' to 3°. The tube draws a spiral of that diameter to find an object that landed just outside the field. A long Shift stops the motion, as for a goto.

Lock

Lock reduces the action menu to **Unlock**. The direction buttons still work. Useful when you hand the controller to someone who should not start a goto.

---

15. Alignment

Alignment computes the transform between the sky and the axes. A two-star alignment uses Taki's method (two measured directions), completed by the nearest proper rotation in the least-squares sense. The result is a 3×3 matrix. It is used for pointing, sync, corrected tracking, and the altitude check. **4 Stars** and **3+3**

Stars, described below, can also measure CH and NP. The result is reported as CH, NP, ID and ME, Wallace's TPOINT signs.

Telescope Action → Align

- **2 Stars**
- **4 Stars**
- **3+3 Stars**
- **2 Stars Mech.** (equatorial mounts)
- **PC Alignment**
- **Mount error**

On an alt-azimuth mount, **2 Stars Mech.** is not offered. **Mount error** stays the last line. Once a model is stored, **Save**, **Clear**, and **Show align. error** are inserted above it.

2 Stars, from home

1. The tube is at the home position. The display reminds you: **The mount must** be at the home position.
2. Choose **Home** when the display asks for the mode.
3. The main unit accepts the start. Pick a star from the list (named stars, above the horizon). The list stays in normal catalog order. Each star shows a suitability mark on the right of its name, from * to ****: more stars mean a better pick for the next alignment measurement. Stars already measured in this session are hidden. The cursor opens on the best remaining candidate; North and South then browse the catalog as usual. On a German equatorial or an equatorial fork a good pick is at a comfortable altitude, well off the meridian, and far from the stars already taken. After the first star is accepted and recentered, the pier side is known: for **4 Stars** and for the first three stars of **3+3 Stars**, the list shows only that pier side (the other side and the strip near the meridian are removed). After the flip in **3+3 Stars**, the list shows only the opposite pier side. On an alt-azimuth mount, including an alt-azimuth fork, a good pick spreads azimuth and altitude, away from the zenith. After the first star, **4 Stars** and the first half of **3+3 Stars** stay in the same part of the sky; after the flip the list keeps only the opposite azimuth.
4. The tube slews. **Slewing to**, then **Recenter**, is shown. If the goto is rejected (limits, below horizon) or you abort the slew (long Shift or another key), the star list reopens on that same star so you can pick again or choose another.
5. Center the star with the direction buttons. A **short** press on Shift accepts the star (**Star added**). To change the recentering speed, a **long** press on Shift alone opens **Set Speed** (Guiding, Slow, Medium, Fast, Max). **Shift + North** or **Shift + South** open the same menu, as outside alignment.
6. Pick the second star. On an equatorial mount prefer one far from the first in hour angle and declination. On an alt-azimuth mount prefer one far in azimuth and altitude. The suitability marks (*...****) and the starting cursor guide that choice; already-used stars are gone from the list. Same recentering: short Shift accepts, long Shift changes speed.
7. On success the model is computed. The hand controller shows **Alignment** / **Success**, then the residual (for **4 Stars** and **3+3 Stars** that is the rigid RMS over all stars, not the old two-star check), then (for those rigid sessions) the head geometry (CH, NP, ID). Each of those screens stays until you press a key. **Save** writes the model to memory. For **4 Stars** and **3+3 Stars**, it also copies the published CH and NP into **Mount error**, so a later **2 Stars** with Mount error On (and the web Mount page) keep those numbers. The Mount error On/Off switch itself is not changed. Without a save the model is lost at the next park or power-off, depending on whether you park: the group wiki reminds you to park at the end of the procedure to keep the result. **Save** in the Align menu does that write explicitly.

Clear forgets the model. The mount returns to the assumption "mechanically correct, starting from home".

2 Stars, from a star

If home is not reachable (the tube is already on the sky):

1. Choose **Star** instead of **Home**.
2. Set the **Side of Pier** (East or West). On a fork or an alt-azimuth mount, set the side that matches the real pose.

3. **Sync** on a centered star. That star becomes the first reference.
4. The procedure continues with the second star, as above.

Mount error during a 2-star alignment

Set **Mount error** before you start, from this menu or from **Telescope Settings** → **Mount**. When it is **On** and at least one of **CH** and **NP** is not zero, the two stars still measure the pole and the axis index. Both numbers are kept, including a zero on the other line. They are not folded into the pole. The mode line on the start screen is **Measures pole** instead of **Alignment**.

4 Stars

From home, on one pier side. The screen says **Same pier side** and **NP**. Confirm with **YES**. Four stars estimate NP. CH is not estimated from a single pier side. The stored **Mount error** values are not used. After the first star is accepted, the star list keeps only that pier side.

The first two stars set the pole. After the third star the model is updated before the slew to the fourth, so a mount that starts a long way off is closer on that last star. The hand controller waits while the update runs. The fourth star closes the session.

3+3 Stars

For a German or fork mount that can change pier side. The screen says **Flip after 3** and **CH + NP**. Confirm with **YES**. Take three stars on one side, flip, then three on the other. After the first star is accepted, the list shows only that pier side; after the flip it shows only the opposite side. After the third star, and after each later star except the last, the model is updated before the next slew. The hand controller waits while that update runs. CH is estimated only when each side really contributed three stars. One star past the flip is not enough, and CH stays unpublished until that split is met. The stored **Mount error** values are not used. The result is shown as CH, NP and ID, in arcseconds, and the pole as MA and ME. The angle to rotate the mount in azimuth is MA divided by the cosine of the latitude.

2 Stars Mech.

Reserved for German equatorial and fork mounts. The same steps, but the calculation is forced to stay consistent with the mount's mechanical pole. Use it when polar alignment is the reference and you want a model that does not twist the pole. The start is still **Home** or **Star**.

From a computer

PC Alignment turns on alignment-by-sync and treats the current pose as home, so the tube must already be there. The computer does the rest. Nothing recenters the star, neither on the hand controller nor from the computer.

For each of two stars:

1. Command a goto. An equatorial goto and an alt-azimuth goto both work.
2. Plate-solve.
3. Sync on the solved right ascension and declination. Sync on those solved coordinates even when they differ from the coordinates sent to the goto. That difference is what the model measures. A sync that still carries the goto coordinates throws the residual away.

The first sync anchors the axes on the solved position. The second builds the model from where the mount stopped and where the solve says the sky is. **Save** writes the model.

A sync after that only corrects the current pointing, for example after another plate solve. It does not clear the model. **PC Alignment** again, or **Clear**, does reset it.

If **Mount error** is **On** and at least one of **CH** and **NP** is not zero, this sync alignment holds both numbers and still measures the pole, the same way as **2 Stars**.

PC Alignment on the hand controller is only that two-star sync. **4 Stars** and **3+3 Stars** from the computer use the actions below.

ASCOM and Alpaca

The Windows ASCOM driver and Alpaca (chapter 24) use the same action names. The tube is at home before **AlignStart**. The program slews, plate-solves, and accepts the solved coordinates. It does not recenter. Sync does not add a star of this session.

Action	Parameters	Role
`AutoAlign`		Same two-star sync as **PC Alignment**. Then two syncs of the solved position.
`AlignStart`	`2`, `m`, `r3`…`r9`	Start from home. `r4` is **4 Stars**. `r6` is **3+3 Stars**. `m` is **2 Stars Mech.**
`AlignAtTarget`	`2`, `m`, `r3`…`r9`	The tube is already on the first star.
`AlignAccept`	`star` or `star,raHours,decDeg`	Record that star against the solved coordinates. Right ascension is hours, declination is degrees.
`AlignSave`		Write the model, the same as **Save**.
`AlignAbort`		Abandon the session.
`AlignPolarDone`		Close **2 Stars Mech.** after the bolt pass.
`AlignStatus`		Stars collected, residual in arcseconds, which of CH, NP and ID were solved (`CPI`, a dash where a term was not), and how many stars lie on each pier side.
`AlignErrors`		CH, NP, ID and the residual in arcseconds, and ME, MA and the two-star error in degrees.

An alt-azimuth slew is fine. The coordinates given to **AlignAccept** are still the solved right ascension and declination.

```
Action("AlignStart", "r6") Action("AlignAccept", "1,12.345678,45.123456") Action("AlignSave", "")
```

Repeat **AlignAccept** for stars 1 through 6, each time with that star's solved coordinates. Alpaca sends the same names to `PUT /api/v1/telescope/0/action`, fields `Action` and `Parameters`.

AlignErrors returns CH, NP, ID and the residual in arcseconds, and ME, MA and the two-star error in degrees, for example `ch=1.500000;np=-2.250000;id=0.000000;me=0.209444;ma=-1.500000;rms=12.500000;ae=0.016667`.

A sync once the model is saved only corrects the current pointing. **AutoAlign** again, **AlignStart** again, or **Clear**, resets the model. While the program is accepting stars, leave the hand controller alone.

When a program instead asks you to recenter each star, the display shows **Remote Align** and the name of the requested star. A short press on Shift accepts the star. A long press on Shift, alone or with North or South, opens **Set Speed**.

Which pair of stars

Pick two bright stars, well above the horizon, separated by at least about forty degrees, not both near the pole and not both near the horizon. One star in the east and one in the west, with different declinations, gives a stable model. Do not accept a star you are not sure of: one catalog mistake is carried across the whole sky.

Show align. error shows the residual angular error of the model. An error of several tens of arcminutes is a reason to repeat the procedure, not a reason to compensate with motor backlash.

What alignment does not fix

It does not replace a wrong gear ratio, a reversed direction, or a wrong clock. If the first goto lands degrees away from the target, go back through chapters 8 to 10 before you start another alignment.

TPOINT names and what TeenAstro really computes

The hand controller and the web page report **CH**, **NP**, **ID**, **ME** and **MA** with Wallace's TPOINT signs. Those names are the usual language for mount errors. TeenAstro does **not** point the sky with the classic linearized TPOINT dial formulas. It uses exact rotations: the head geometry (collimation, non-perpendicularity, index) and a real orientation of the polar axis in space.

TPOINT was built to fit small residuals in hour angle and declination on a few stars. Its geometric terms are convenient corrections for least-squares, valid mainly for **small** (almost infinitesimal) errors. **TeenAstro** asks a different question: where does the polar axis point, and how is the head built.

```
TPOINT: which ΔHA, ΔDec patches reduce residual on my stars? TeenAstro: what is the
orientation of the head and polar axis in space?
```

Classic TPOINT (first-order, on the dials), with (H) = hour angle and (\delta) = declination:

[\begin{aligned} \Delta H &= \mathrm{IH} + \mathrm{CH}\,\sec\delta + \mathrm{NP}\,\tan\delta + \mathrm{MA}\,\cos H\,\tan\delta - \mathrm{ME}\,\sin H\,\tan\delta \[0.4em] \Delta\delta &= \mathrm{ID} - \mathrm{ME}\,\cos H + \mathrm{MA}\,\sin H \end{aligned}]

TeenAstro's pole is a direction in the local horizontal frame (a plumbed mount). If (d_{\mathrm{Az}}) and (d_{\mathrm{Alt}}) are the azimuth and altitude tilts of the mechanical polar axis, and (\phi) is the latitude:

[\mathrm{MA} = d_{\mathrm{Az}}\cos\phi,\qquad \mathrm{ME} = -d_{\mathrm{Alt}}]

The angle you turn the mount in azimuth to correct the pole is therefore (\mathrm{MA}/\cos\phi), as already stated for **3+3 Stars**. EEPROM stores that azimuth tilt, not MA itself.

```
true celestial pole * /| / | ME (altitude of the mechanical pole) / | /___|_______ horizon /
dAz * mechanical polar axis
```

Term	Meaning on the display	In TeenAstro
CH	collimation (east–west)	head cone, same sign
NP	non-perpendicularity	head, opposite internal sign
ID	declination index	head index
ME	polar elevation	altitude of the polar axis (sign as above)
MA	polar azimuth	(d_{\mathrm{Az}}\cos\phi) on the polar axis

CH, **NP**, **ID** and **ME** match the usual TPOINT idea closely enough for night use. **MA** is the exception: the classic TPOINT formula for MA is a dial patch (\Delta H = \mathrm{MA}\,\cos H\,\tan\delta), (\Delta\delta = \mathrm{MA}\,\sin H). That is **not** the same as tilting the polar axis in azimuth. TeenAstro's displayed MA is always the plumbed pole tilt. Software that expects the old tan/sec MA operator is speaking a different language.

```
Physical MA (TeenAstro) Classic TPOINT MA formula ---------------------- ------------------
------ One polar-axis direction Linear ΔH, Δδ on the dials in space (valid at large angle)
(small-angle approximation)
```

For the full formulas, sketches and the cross-check against OnStep, see the math note alignment.md (section *Extra: TPOINT vs TeenAstro geometry*).

---

16. German equatorial mount

The tube can be on either side of the pier. The display shows the side. A goto that would cross the meridian past the limits starts a **Flip**: the tube changes side, while the object's right ascension and declination stay the same.

Goto → Flip requests it at once. If the arrival pose is outside the limits, the display shows **Flip / Not possible**.

Side of Pier forces East or West when the mount has lost that information (clutches loosened, tube moved by hand). The menu then asks for a sync on a target. Without it, the displayed side and the sky do not agree.

Meridian limits

Limits → German Equatorial

- **Meridian E** and **Meridian W**: from −45° to +45°. They allow tracking a little past the meridian before the flip, so a exposure is not cut at the crossing.
- **Under Pole**: maximum hour angle, from 9 h to 12 h. It keeps the tube or the counterweight bar out of the tripod when you track an object under the pole. An equatorial fork uses the same hour angle. A goto past it changes pier side: declination goes through 90° and the tube looks under Polaris. The fork does not flip at the meridian.

Set these values for your mechanics (tube length, tripod height), not for the catalog. A refused object is announced as **Outside Limits**, **Below Horizon**, **Close to Zenith**, or a meridian limit.

---

17. Alt-azimuth mount

The base must be horizontal. A bubble level on the base is part of the setup: a two-star alignment absorbs a residual, not a base tilted by several degrees.

Default home: tube horizontal, toward the south. You can define another one with **Set Home** once the tube is in the pose you want.

Tracking corrects both axes all the time, because neither axis is parallel to the axis of the Earth. Time and site are mandatory: without them, field rotation and tracking are wrong even if the tube was synced on an object.

Under Pole does not apply to this mount type. The useful limits are the horizon, the zenith, and the axis stops.

Near the zenith, a small move on the sky needs a large move in azimuth. The **Overhead Limit** (next chapter) keeps the tube out of that zone.

---

18. Limits

Mount → Limits

Line	Meaning	Entry range
Horizon	Minimum altitude of a goto	−10° to +20°
Overhead	Maximum altitude	60° to 91°
Axis	Mechanical stops of both axes	in degrees, after the current position is shown
German Equatorial	Meridian and under-pole	see chapter 16

Horizon at 0° refuses anything below the horizon. A negative value allows a slight pass below the horizon, for example from a site with a clear drop. A positive value keeps the tube above a wall or a hedge.

Axis first shows the position, then **Axis1 Min.**, **Axis1 Max.**, **Axis2 Min.**, **Axis2 Max.** Move the tube by hand, or slowly on the motors, against each real stop, read the angle, and pull the limit back by a degree or two so the mechanics are not hit.

A refused goto is not a fault. Read the message: **Below Horizon**, **Close to Zenith**, **Outside Limits**, declination limit, or azimuth limit.

---

19. Gear check

Telescope Action → Gear Check compares the stored gear ratio with the real motion.

The firmware asks for a sync on a star A, then a goto to B (a large move mostly on axis 1) and to C (a large move mostly on axis 2). For an equatorial mount it suggests a large hour angle, then a large declination. For an alt-azimuth mount, a large azimuth, then a large altitude.

At each arrival:

1. Wait for the end of the slew (**Waiting slew...**). A long Shift aborts.
2. Recenter the star.
3. Short press on Shift to confirm.

The display shows the error of each step, in arcminutes, and the measured gear. The formula is: measured gear = stored gear × true move / commanded move, where the true move is the commanded move plus the recentering error. A shortfall (you still had to push the star further) raises the gear; an overshoot lowers it. A commanded move of less than 5° is rejected: the scale would be too poor.

If the two axes give inconsistent ratios, the display reports **Axes similar**: the chosen stars did not separate the axes enough. Start again with targets farther apart.

Copy the measured gear into **Motor** → **Gear** only if the rotation direction is already correct and you recentered carefully. A poorly centered star becomes a gear error directly.

---

20. Encoders and push-to

Encoders measure the real axis position. They are used for push-to (you push the tube, the display shows how to reach the target) and they can resync the motors.

Mount → **Encoders** → **Enable**, then reboot.

Then:

Line	Role
Auto Sync	Automatic resync of the motor onto the encoder
Calibration	Calibration on a star
Pulse per deg E1 / E2	Encoder resolution
Reverse E1 / E2	Encoder direction
Disable	Turns the encoders off, with a reboot

Pulse per deg runs from 0.50 to 3600 pulses per degree.

Auto Sync: Off, 60', 30', 15', 8', 4', 2', or On. The motor is resynced onto the encoder when the error exceeds the threshold, and only while not slewing. **On** resyncs continuously at the finest tolerance the firmware provides. **Off** leaves the encoders informational.

Calibration

1. **Star**: starts the calibration.
2. Point at the reference star and center it as the display asks.
3. **Complete** stores it. **Cancel** aborts.

Each encoder's **Direct** / **Reverse** direction is set like the motors: if push-to moves away when you push in the direction shown, reverse that encoder.

Push-to

With encoders enabled, **Telescope Action** contains **Pushto**. Choose the target as for a goto. The display switches to the distance page. Move the tube by hand until the error is gone. A short press on Shift then offers to sync the motors to the encoders.

Without encoders, that entry does not exist: the action menu is the one in chapter 14, without the Pushto line.

---

21. Autoguider port

The ST-4 port on the main unit receives the four directions from an autoguider or a guide camera. The speed used is the **Guiding Speed** of chapter 12.

Tracking must be running. An ST-4 correction is added to tracking; it does not replace it. The guiding state (pulse, ST-4, recenter) is visible to connected software, and the motion is visible at the eyepiece.

Set the guiding speed so that the star moves clearly on a one-second pulse, without crossing the whole sensor. 0.5× sidereal is a usual starting point. Go toward 0.3× at long focal length, and toward 0.8× if the corrections have no effect.

22. Focuser

If no focuser is connected, Shift + F or Shift + f shows **Focuser** / **Not Connected**.

Focuser Action (Shift + f): named positions already stored, **Goto**, **Sync**, **Park**, **Lock**. Choosing a named position sends the focuser there. **Sync** declares that the current position is the reference. **Park** sends the focuser to its park position.

Focuser Settings (Shift + F):

- **Config**: display, park position, maximum position, manual and goto speeds, accelerations.
- **Motor**: resolution, direction, steps per rotation, microsteps (4, 8, 16, 32, 64, 128), current.
- **Focuser Info**: version, reboot, factory reset of the focuser alone.

Locking the focuser keeps an accidental press from losing focus during an exposure.

23. Plate solving

Two ways to plate-solve: a TeenAstro eFinder on the focuser or GNSS port, or the phone camera in the Android app. The phone does not need the eFinder. Either one can sync the mount, and either one can sync and recenter after a goto.

The finder does not look along the telescope. The difference is two rotations in the telescope frame, the **Hz cone** and the **Ve cone**. Hz is the horizontal cone and Ve is the vertical cone. They belong to the tube, or to the phone clamped on the tube. Measure them once, on a bright star. Later solves follow the telescope.

eFinder on the hand controller

eFinder appears under **Telescope Settings** only after the main unit has found the camera.

Solve reports the field and does not change the model. **Solve & Sync** writes that position into the mount.

Aim. Go to a bright star, center it in the telescope, and sync. Then **Aim**. The hand controller asks for that bright star, already centered and synced, and shows **Hz Cone** and **Ve Cone** in arcminutes. Aim fails when the tube is too close to the pole on an equatorial mount, or too close to the zenith on an alt-azimuth mount. Choose another star.

Calibrate is a different step. It places the brightest star against the center of the camera. It does not measure the Hz cone or the Ve cone.

GoTo++. After a goto, the eFinder solves, syncs, and if the telescope is still farther from the target than the threshold, slews back to that target once and solves again. The **GoTo++** line turns this on or off and sets the threshold in arcminutes.

Align starts an alignment from plate solves. **Align** takes the first star. Slew to a second part of the sky and use **Solve & Sync**. The second accepted star closes the model, as in chapter 15, without centering each star by hand. **Align** does not erase a model that is already stored. Clear that model first if you want a new one.

Exposure and **Gain** set the camera. **Info** shows the eFinder version.

Phone finder

On Android, open **Phone finder** from the dashboard or from the alignment screen. Leave that screen open: the camera has to stay on.

Set **Field** to the angle the phone lens covers, from 15° to 90°. The default is 60°. The catalog is bright stars, down to magnitude 4.5, so a narrow eyepiece field will not match. Clamp the phone to the tube.

Solve shows the telescope axis once the cones are known, and the camera axis while they are still zero. **Solve & Sync** sends that axis to the mount.

Aim is the same procedure as on the hand controller. Center a bright star in the telescope and sync, then capture. The screen shows **Hz Cone** and **Ve Cone**. **Clear cones** forgets them.

Auto sync. Turn it on and leave the Phone finder screen open. When a goto finishes, the phone solves, syncs, and recenters once if the telescope is farther from the target than **Recenter**. The limit runs from 2 to 60 arcminutes and starts at 2. The switch and the limit stay on the phone.

The phone does not run **GoTo++** inside the main unit, and it does not start **Align**. It syncs once and, when the miss is past the limit, recenters once.

---

24. Wi-Fi, phone, computer

Telescope Settings → Wifi

- **Turn Wifi on** or **Turn Wifi off**
- **Show Password**
- **Select Mode**: up to three station networks, plus the hand controller's access point
- **Show IP**
- **Reset to Factory** of the hand controller's Wi-Fi configuration

Turning Wi-Fi on or off requires a reboot.

In access-point mode, the phone or the computer joins the hand controller's network, with the password shown on the display. In station mode, the hand controller joins your router: the address is the one **Show IP** reports, not a fixed address.

The TCP bridge listens on port **9999** and carries the same dialogue as the main unit's USB cable. The TeenAstro app, the ASCOM driver, and planetarium software connect there.

The app provides the dashboard, the planetarium, goto, and alignment. The ASCOM driver opens TeenAstro to Stellarium, Cartes du Ciel, NINA, and other ASCOM clients on Windows.

A web page on the interface lets you enter networks and the password without using the hand controller. The address is the one shown by **Show IP**. The mount page on that site has **Mount error** above refraction: see chapter 7.

ASCOM Alpaca

With a **Lolin S3 Mini** (ESP32-S3) in the hand controller, TeenAstro also serves **ASCOM Alpaca** over Wi-Fi: REST/JSON on TCP **11111**, discovery on UDP **32227**. NINA, Stellarium, KStars, ASCOM Remote, and other Alpaca clients can find the mount on the LAN without the Windows COM driver or port 9999.

Required:

1. Hand controller fitted with a Lolin S3 Mini (not the Wemos D1 Mini / ESP8266 — that image has no room for Alpaca). Swap steps: group wiki Upgrade SHC to ESP32-S3.
2. S3 firmware (`TeenAstroSHC_*_S3_English.bin`, or French / German).
3. Wi-Fi on, computer or phone on the same network, address from **Show IP** (or **192.168.0.1** in access-point mode).

The web UI stays on port **80**; the classic ASCOM driver and the TCP bridge on **9999** remain available. Plate-solve alignment from either driver is in chapter 15, under **From a computer**. Details: ASCOM Alpaca on the group wiki.

SkySafari

You need SkySafari Plus or Pro, the edition that drives a telescope. The phone and the hand controller must be on the same network.

In access-point mode, as shipped: the network name is **TeenAstro**, the address is **192.168.0.1**, the port is **9999**. The factory password is `password`; **Show Password** displays the one actually stored, if it was changed. In station mode the phone is on your router and the address is the one from **Show IP**, still on port 9999.

In SkySafari: Settings, Telescope, Setup.

Setting	Value
Mount type	Equatorial GoTo, or Alt-Az GoTo, matching the mount
Scope type	Meade LX-200 Classic
Connection	Wi-Fi. Not the SkyFi mode
Address	192.168.0.1 in access-point mode, otherwise the **Show IP** address
Port	9999

Set Time & Location, if offered, sends the phone's clock, and the site only while the mount is at home or parked. If time and site are already right on the hand controller, leave that box off: the phone would overwrite the site. If you do want the phone's GPS, park first, or be at home, then enable the box and connect.

Connect. The chart should show the telescope crosshair. Tap an object, then Goto. The hand controller shows the slew icon. One goto at a time: SkySafari's and the hand controller's interrupt each other.

Recentering is still done with the buttons, slow speed then guiding. Sync can come from SkySafari or from the hand controller, on the same object. The two-star alignment of chapter 15 stays simpler on the hand controller; SkySafari is for choosing objects afterwards.

While a computer commands the mount, the hand controller remains usable. Avoid starting two gotos at once: the second one interrupts the first.

Firmware update

Install **TeenAstroUploader** from the MSI (`TeenAstroUploader.msi`). The group wiki page TeenAstroUploader for Windows has the same steps with pictures.

The window is small and classic: **Firmware Version**, **Stable** / **Latest**, **Download!**, **Open Folder**, then three tabs — **Telescope**, **Focuser**, **Hand controler**.

1. Choose the firmware version and press **Download!** once (Stable or Latest). Wait until the progress bar finishes.
2. USB cable on the device you flash only (main unit, or focuser, or hand-controller Wemos), mount powered. The first time, Windows installs the Teensy (or CH340) driver.
3. On **Telescope** or **Focuser**: - **Upload!** flashes the selected PCB only. Use this on firmware older than 1.5, or when you already saved the settings yourself. - **Auto!** (firmware **1.5 or newer**) detects the board, backs up parameters to a JSON file, flashes, restores after reboot, then **verifies** the values. TeenAstro stores **two mounts** (index 0 and 1); Auto works on the active mount. Older than 1.5 is refused: those release branches are no longer supported for backup/restore. - **EEPROM** opens the parameter editor (same groups as the Webserver: Mount, Motors, Speed, Limits, Encoders, Site, Tracking). Read / Write / Save / Load. After Write, the tool re-reads the device to check the values. Use **Activate (reboot)** to switch between the two stored mounts.
4. On **Hand controler**: select language and COM port, then **Upload over COM!**; or enter the IP from **Show IP** and **Upload over WIFI!**.

Jürgen Goldan's TeenAstroConfig remains useful for older workflows; for 1.5+ the uploader **EEPROM** and **Auto!** cover the same need.

The hand controller and the main unit must come from the same release. Otherwise the version-error screen returns at startup.

Main Unit Info → **Show Version** shows the firmware name and date. **Reboot** restarts the unit. **Reset to Factory** erases memory after a NO / YES confirmation.

---

25. Visitor mode

SHC Settings → **Rights**: **Administrator** or **Visitor**.

In visitor mode, **Telescope Settings** contains only **Rights**. Mechanical, time, and Wi-Fi settings are hidden. Night-time actions (goto, sync, tracking, speeds) remain available. This is the mode to leave on a shared

mount or during a demonstration.

The choice is stored in the hand controller. To return to administrator: Shift + West, **Rights**, **Administrator**.

---

26. An observing night

The example is a late-summer evening around 45° north: Vega and Altair are high, M13 is still up, M31 is rising. On another night, keep the same actions and change the names. Filters are in chapter 14, SkySafari in chapter 24.

Telling that the battery is empty

The hand controller has no gauge. Nothing on the screen says "battery low". You see it in the behaviour, and with a check before the night.

Before power-on, a voltmeter on the battery is the only reliable number. The exact threshold depends on the chemistry (lead or lithium) and on the board: the group wiki gives the voltage for your version. A battery that holds at rest and sags as soon as a motor accelerates is already too weak.

During the session, a failing battery looks like this, while the cables have not moved:

- the display dims, flickers, or the hand controller restarts on its own at the logo;
- **Not Connected**, then a reboot, without anyone touching the cable;
- a goto at Max speed stalls, the motor growls, the tube stops short of the object, and the same move at slow speed succeeds;
- Wi-Fi disappears and **Show IP** no longer answers until the next start;
- tracking stops and the coordinates no longer match the sky, because the main unit restarted and lost its place.

These are not an empty battery: a black screen that Shift wakes is the sleep timeout; **ERROR** / **version** is firmware that does not match.

If the signs appear during a goto, stop the motion (long Shift). Park only if the motors still answer. Recharge or swap the battery before going on: lost steps make the pointing wrong until a new sync.

Starting up

1. Tube at the park position, or at home if you have not defined a park yet. Connect, power on, let the logo, the versions, and the time go by.
2. Check the time and the site (**Time & Site**). Without GNSS, those are the values you entered.
3. If the tracking icon is absent: **Telescope Action** → **Tracking** → **Start Tracking**.
4. If the action menu offers only **Unpark**, the mount is parked. Unpark. The park icon must go away.

Align on two stars

If a model already exists and the polar alignment has not moved: **Goto** → **Catalogs** → **Bright Stars**, choose Vega, center at slow speed, and sync only if the star entered the field. Then go on to the objects.

Otherwise, a two-star alignment from home (chapter 15), with two named stars, high, and far apart:

1. First star: Vega, in Lyra (look for a high **** mark). The tube moves, the screen says **Slewing to** then **Recenter**. Center it. Short press on Shift: **Star added**. Long Shift opens **Set Speed** if you need a slower step.
2. Second star: Altair, in Aquila. It is far from Vega in hour angle; Vega is no longer in the list. Same recentering, short Shift to accept.
3. Read the success and error screens (each waits for a key). **Save**. The alignment icon confirms that the model is stored.

A pair that is too close, Vega and Deneb for example, gives a weak model. Chapter 15 explains how to choose.

First object: M13

M13 is the Hercules globular cluster, easy in a finder once the model is in place.

1. **Goto** → **Catalogs** → **Filters**.
2. **Above Horizon**, then **Above 30 deg.** M13 should be well up; if the filter removes it, that is the right outcome.
3. **Constellation**, then **Her**.
4. **Type**, then **Globular Clstr**.
5. Back to the catalogs, open Messier. The title becomes **!Goto Messier!**: the filters are working. Scroll to M13. F starts the goto.
6. On arrival, **Slow** in the finder, **Guiding** in the eyepiece. Sync only if you are sure the cluster is centered. A sync on the wrong object shifts the whole sky.

If the screen says **No Object**, M13 is below 30° or the type does not match. **Reset Filters**, then try again with **Above 10 deg.**

Second object: M31

M31, the Andromeda galaxy, is not in Hercules. Leave the constellation filter unchanged and it will not appear.

1. **Filters** → **Constellation** → **And**.
2. **Type** → **Galaxy**. Leave the altitude at 30° if Andromeda is already up, otherwise drop to **Above 10 deg.**
3. Messier, M31, F.
4. The field is wide: slow speed, and the finder rather than high power, to pick it up.

For a planetary nebula in the same part of the sky as Vega: **Constellation** → **Lyr**, **Type** → **Planetary Nebula**, then M57.

The Moon, Jupiter, or Saturn go through **Goto** → **Solar System**, with no filter. The Sun only with a full-aperture filter on the tube. The controller does not check that.

End of the night

Goto → **Park**. Wait for the park icon, not merely for the motors to stop. Then switch the power off. An interrupted park leaves the position wrong at the next start.

If you loosened the clutches during the night, do at least a sync again, and an alignment if the tube moved a long way without the encoders following.

---

27. When something is wrong

What you see	Where to look
ERROR / **version**	Hand controller and main unit from different releases
Not Connected, then a reboot	Hand-controller cable, a battery sagging under load, or the main unit off
The display dims, the hand controller restarts at the logo, a Max-speed goto stalls	Battery empty or too weak under load. The hand controller does not show voltage. Chapter 25
Black screen, Shift wakes it	Sleep timeout, not an empty battery
The buttons move the wrong way	Chapter 8, rotation direction
Goto is always off by a factor of two or more	Gear ratio or step count. Run a gear check
Goto is good near the sync star and worse farther away	No alignment, or the wrong star. Wrong time or longitude
Below Horizon or **Outside Limits** on an object that is visible	Horizon limit, wrong site, or wrong pier side
Close to Zenith	Overhead limit, or the object really is too high for the mechanics
The motor stalls on a goto	High current too low, Silent, acceleration too sharp, max speed too high

What you see	Where to look
The motor heats while idle	Low current too high
The star jumps at every button reversal	Backlash too large, or backlash at zero while the mechanics have some
Alignment fails immediately	The mount was not at home, or the first sync was rejected
Flip / **Not possible**	The arrival pose would cross a meridian or axis limit
Push-to moves away from the target	Encoder direction reversed, or wrong pulses per degree
Wi-Fi shows no address	Wi-Fi off, wrong mode, or the reboot has not happened yet

Before you erase memory: write the parameters down, or reopen the TeenAstroConfig backup. A factory reset is the last resort, not the first.

Appendix A. Menu tree

Lines in parentheses appear only in the case indicated.

Shift + East — Telescope Action

- (if parked) Unpark
- Goto — Catalogs, Solar System, Coordinates, User Defined, Home, Park, Flip
- (encoders) Pushto — Catalogs, Solar System, Coordinates, User Defined
- Sync — same as goto, without Flip
- Align — 2 Stars, 4 Stars, 3+3 Stars, (equatorial) 2 Stars Mech., PC Alignment, Mount error; once a model is stored, also Save, Clear, Show align. error (Mount error stays last)
- Gear Check
- Tracking
- Side of Pier
- Save RADEC
- Lock
- Spiral

Shift + North — Set Speed: Guiding, Slow, Medium, Fast, Max

Shift + West — Telescope Settings (administrator)

- Hand Controller — Rights, Display, Button Speed, Ergonomics, Reset
- Time & Site — Time (Clock, Time Zone, Date, GNSS Time), Site, Sync to GNSS
- Park & Home — Set Park, Set Home, Reset Home
- Mount — Mount, Mount type, Motors, Encoders, Limits, Mount error, Refraction, Reticule
- Main Unit Info — Show Version, Reboot, Reset to Factory
- Wifi
- (eFinder connected) eFinder — Solve & Sync, Solve, Align, GoTo++, Calibrate, Aim, Exposure, Info

Motors → Motor 1 or 2: Show Settings, Rotation, Gear, Steps per Rot., MicroStep, Backlash, Backlash Speed, Low Curr., High Curr., Silent

---

Appendix B. Greek letters

Star catalogs use Bayer letters. The English firmware names the catalog "Bright Stars".

Letter	Name	Letter	Name
α	alpha	ν	nu
β	beta	ξ	xi
γ	gamma	ο	omicron
δ	delta	π	pi
ε	epsilon	ρ	rho
ζ	zeta	σ	sigma

Letter	Name	Letter	Name
η	eta	τ	tau
θ	theta	υ	upsilon
ι	iota	φ	phi
κ	kappa	χ	chi
λ	lambda	ψ	psi
μ	mu	ω	omega

A few proper names that often appear in that catalog: Sirius, Canopus, Arcturus, Vega, Capella, Rigel, Procyon, Betelgeuse, Altair, Aldebaran, Spica, Antares, Pollux, Deneb, Regulus.

For alignment, a named star high in the sky is worth more than a faint star you are not sure of.

Appendix C. Computing the gear ratio

Gear = (wheel teeth) × (ratio of every reducer or pulley ahead of it).

Arrangement	Calculation	Value to enter
Worm, 360-tooth wheel, motor direct	360 × 1	360
180-tooth wheel, 16/8 pulley	180 × 2	360
144-tooth wheel, 10:1 reducer	144 × 10	1440
360-tooth ring, stages 3:1 and 2:1	360 × 3 × 2	2160

Steps per rotation are those of the bare motor (200 for 1.8°, 400 for 0.9°), not the product with the microsteps. Microsteps are set separately.

Approximate sky resolution, in arcseconds per microstep:

360 × 3600 / (gear × steps per rotation × microsteps)

Example: gear 360, 200-step motor, 16 microsteps → 11.25" per microstep. That is the order of magnitude of the smallest move, before driver interpolation. It is not the pointing accuracy, which depends on backlash, flexure, and alignment.

Appendix D. Where to go next

- Group and wiki: https://groups.io/g/TeenAstro/wiki/home. First run, illustrated menus, boards, the backup tool, and the flashing procedure are there.
- Technical documentation in the repository: docs/README.md (architecture, tracking, protocol). It is for people who change the software, not for running a night.
- Alignment math, including TPOINT versus TeenAstro geometry: docs/math/alignment.md. Chapter 15 of this manual has the field summary.
- FS2 manual, for the original approach: anleit_f.pdf (French) and the German edition on the same site, Astro-Electronic, Michael Koch.

Boards, supply voltages, and motor harnesses are not the same from one version to the next. For wiring, start from your board's page in the wiki, then come back here for use.

Appendix E. Changes since version 1.6.0

Firmware 1.6.0 is the first release of this generation. The ASCOM driver readme, installed with the driver, describes the jump from 1.5 to 1.6. This appendix lists what changed afterwards, through firmware 1.6.9 and the matching ASCOM driver, then the computer alignment actions of the current drivers.

1.6.1

- During a goto, the hand controller can tell an equatorial slew, an alt-azimuth slew, and a meridian flip apart.

- The Wi-Fi setup page no longer overflows, and the web server is more stable.
- The focuser temperature sensor starts reliably. The ASCOM focuser reports a missing sensor without crashing the client.
- The ASCOM setup can read and write the telescope and focuser settings stored in the mount.

1.6.2 to 1.6.4

- Catalog star names are stored and shown correctly. Long hand-controller texts were shortened so they fit the display.
- Park and home gotos complete. If a computer starts an alignment, the hand controller shows that alignment.
- The hand controller shows which Wi-Fi mode is active, and the access point comes up reliably.
- Tracking rates travel at full precision. MoveAxis uses the mount's effective maximum slew rate.
- Guiding is reported in the mount status, so a client can see that a guide pulse is in progress.
- ASCOM goto coordinates round-trip more accurately. A dropped serial or Wi-Fi link is retried. Motor reverse uses the documented settings.
- The TeenAstro app gained the planetarium, night view, J2000 and JNow, and catalog filters.

1.6.5

- The main unit and the hand controller were brought to the same version number. No new night-time action.

1.6.6

- A meridian-flip goto is allowed to finish, tracking starts again afterwards, and the meridian direction matches the mount.
- Mechanical polar alignment (**2 Stars Mech.**) is its own procedure and is closed with its own confirmation.
- Wi-Fi in station mode keeps the link. Saving a setting on the web page returns to that page.
- The hand controller is less likely to crash during a long session.

1.6.7

- Gear check on the hand controller, for an equatorial or an alt-azimuth mount, shows the measured ratio.
- The under-pole limit is corrected. A goto that would cross a limit is refused. Aborting an alignment restores the previous model.
- Hour angle stays in a fixed range, including west of the meridian, so a computer does not reject the reply.
- GNSS accepts a time fix only when the positions agree, and a quiet sync does not disturb the session.
- Catalog scrolling is faster, and a semicolon inside a name no longer corrupts the list. Configuration pages load faster. Web forms can be submitted again.
- ASCOM can find home.

1.6.9

- Stopping MoveAxis slows down with the configured acceleration. Tracking continues on the axis that is not being moved. A rate of zero always stops. A rate above the maximum is refused.
- SharpCap and the other clients show the MoveAxis speeds again, including 0.25°/s and 0.5°/s. The driver uses sidereal-rate multiples, the same unit as driver 1.5.
- ASCOM rates, coordinates, and port numbers use a fixed decimal point, so a French or German Windows does not change the commands.
- The hand controller guides **2 Stars**, **4 Stars**, and **3+3 Stars**. Four stars on one pier side measure NP. 3+3 measures CH and NP only when each pier side has three stars. The reported numbers are Wallace's CH, NP, ID and ME.
- After the first accepted star in **4 Stars** or **3+3 Stars**, the hand-controller star list shows only the correct pier side. After the 3+3 flip it shows only the opposite side.
- **4 Stars** and **3+3 Stars** update the model after the third star, and after each later star before the last. The next slew uses the stars already centered. The hand controller waits while that calculation runs.

- **Mount error**, above Refraction, holds a known CH and NP during a two-star alignment and during a plate-solve sync.
- **Save** after **4 Stars** or **3+3 Stars** also copies published CH and NP into Mount error, so the web Mount page and a later 2-star with Mount error On see those values (not 0.000°).
- The firmware uploader **Auto!** button (Telescope / Focuser) backs up parameters, detects the board, flashes, restores, and verifies (firmware 1.5+). **EEPROM** edits the full parameter set (two mounts on the MainUnit). **Upload!** still flashes without touching settings.
- An eFinder on the focuser or GNSS port adds **eFinder** to Telescope Settings, and the letters **eF** on the status row, only while the camera is found. **Aim** stores the Hz cone and the Ve cone in the telescope frame. **GoTo++** syncs after a goto and recenters once past the threshold. Chapter 23.
- The Android app **Phone finder** does the same solve, aim, sync, and one recenter from the phone camera. No eFinder is required.

Alignment from ASCOM and Alpaca

- The Windows driver and Alpaca share one set of alignment actions. **AutoAlign** is **PC Alignment**: two plate-solve syncs, no recentering. **AlignStart** with `r4` or `r6` starts **4 Stars** or **3+3 Stars** from the computer. **AlignAccept** records each solved star. **AlignSave** stores the model. **AlignErrors** reads CH, NP, ID, ME, MA, the residual and the two-star error. Chapter 15 has the sequence. A sync after the model is saved only corrects the current pointing.